

9	(a)	(i)	Explain what is meant by the <i>internal energy of an ideal gas</i> .
			It is the sum of random distribution of kinetic energies of the gas molecules B[1]
		(ii)	Mean random translational kinetic energy of ideal gas molecules = $\frac{3}{2}kT$ For a system containing N molecules of ideal gas, the internal energy = total random kinetic energy = $N \times \frac{3}{2}kT = \frac{3}{2}NkT$ M[1] Since $pV = NkT$ Internal energy = $\frac{3}{2}pV$ A[1]
	(b)	(i)	The molar mass of carbon dioxide is 44.0 g. Show that the mass of one molecule of carbon dioxide is about 7.3×10^{-26} kg. Mass of one molecule = $\frac{44.0 \times 10^{-3}}{6.02 \times 10^{23}}$ M[1] = 7.31×10^{-26} kg
		(ii)	The surface temperature of the planet Mercury is 467 °C. Determine the root-mean-square speed of one molecule of carbon dioxide at this temperature. Average k.e. per molecule: $\frac{1}{2}mc_{rms}^2 = \frac{3}{2}kT$ $\frac{1}{2}(7.31 \times 10^{-26})c_{rms}^2 = \frac{3}{2}(1.38 \times 10^{-23})(467 + 273)$ C[1] $c_{rms}^2 = 419,097$ C[1] $c_{rms} = 647 \text{ ms}^{-1}$ A[1]
		(iii)	The root mean square speed of one carbon dioxide molecule is less than the escape speed. M[1] Hence, an atmosphere of carbon dioxide can exist on Mercury's surface. A[1]

9	(c)	(i)	State the first law of thermodynamics. The <u>increase in internal energy</u> B[1] of a system is equal to the <u>sum of heat supplied and work done on the system.</u> B[1]
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		(ii)	In the pV diagram shown below (not to scale), an ideal gas expands isothermally from point A to point B and then adiabatically from point B to point C where its volume is $24.0 \times 10^{-3} \text{ m}^3$. An isobaric compression brings it to point D where its volume is $10.0 \times 10^{-3} \text{ m}^3$. An isochoric process returns the gas to point A.
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		(1)	Show that the pressure of the gas at point B is $875 \times 10^3 \text{ Pa}$. $p_B V_B = p_A V_A \quad p_B = \frac{p_A V_A}{V_B} = \frac{(1400 \times 10^3)(0.010)}{0.016} \quad \text{M[1]}$ $= 875 \text{ kPa}$
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		(2)	Complete the following table. <table border="1"> <thead> <tr> <th></th><th>q / joules</th><th>w / joules</th><th>$\Delta U / \text{joules}$</th></tr> </thead> <tbody> <tr> <td>A $\rightarrow$ B</td><td>6580 [1]</td><td>- 6580</td><td>0</td></tr> <tr> <td>B $\rightarrow$ C</td><td>0 [1]</td><td>- 3140</td><td>- 3140</td></tr> <tr> <td>C $\rightarrow$ D</td><td>-17340</td><td>6940 [1]</td><td>- 10 400</td></tr> </tbody> </table>		q / joules	w / joules	$\Delta U / \text{joules}$	A $\rightarrow$ B	6580 [1]	- 6580	0	B $\rightarrow$ C	0 [1]	- 3140	- 3140	C $\rightarrow$ D	-17340	6940 [1]	- 10 400
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[3]

		(3)	$\Delta U_{D \text{ to } A} - 3140 - 10\,400 = 0 \quad \text{C[1]}$ $\Delta U_{D \text{ to } A} = 13\,540 \text{ J} \quad \text{A[1]}$
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			or $\frac{3}{2}(1400 - 496) \times 10^3 \times 10 \times 10^{-3} = 13\,560 \text{ J}$
		(iii)	- Gas molecules <u>transfer kinetic energy</u> to the <u>receding piston</u> during collision M[1] - Resulting in a decrease in the average kinetic energy of the gas molecules A[1] - Since average (random) kinetic energy of the gas molecules is proportional to (thermodynamic) temperature of the gas M[1] - Gas temperature decreases A[0]